# matheus hoffmann

## the aquatic prometheus

sr. software engineer and jr. researcher, with 8 years of experience building scalable solutions, mainly for payment processing and fraud-fighting. my research topics are wave modelling, game theory and self-improving apps. i work in the open, and i'm open to commercial offers.

mhoffmannfs@gmail.com · linkedin · github · the site · ways of working

## the arc

| | |
|---|---|
| 2026 | **marola** and **Ouroboric** (BR): open research, with friends. |
| 2025–2026 | **ITV** (UK): consulting services, for one year. |
| 2023–2024 | **Signifyd** (US): senior software and ml engineer, fraud-fighting. in New York for the work. |
| 2021–2022 | **Elemeno AI** (US): lead mlops, for Americanas (BR). |
| 2020 | **Broad** (YC W21): senior software engineer, a digital bank. |
| 2017–2019 | **Stone** (BR, NASDAQ): software engineer, payments. joined before the IPO. |
| 2016–2017 | **LIOC**: the ocean instrumentation lab of UFRJ. |

the whole story, with the knowledge graph of the works, is on the site.

## the works, each with a DOI

**marola** 10.5281/zenodo.23224155
a citizen-science ocean intelligence layer for the brazilian coast: "can i swim tomorrow, and when?". scala 3, rag with citations, designed in public (MIPs).

**ww3-gpu** 10.5281/zenodo.23221351
a kernel of NOAA's WAVEWATCH III wave model, ported to c++ and kokkos. it matches the fortran bit for bit on serial, openmp and cuda.

**marola-bench** 10.5281/zenodo.23352633
a proposal: a WeatherBench for brazil, scored against buoys.

**picasso** 10.5281/zenodo.23311208
my nix labs: flakes that pin every toolchain the other works build with.

**closed source: the god that failed**
10.5281/zenodo.23333656
a position paper on reverse engineering and the ethics of citation in the age of coding agents. not peer reviewed.

**self-improving apps**
writing and doing research for ouroboric-dev: open factories that improve themselves.

## how i work

async and remote first, with 4h of overlap a day at most. no live coding: a code interview, or a talk about past projects. money first. references on request.

## certs

**AWS certified 4x**: Cloud Practitioner, Machine Learning Specialty, Machine Learning Engineer Associate, Data Engineer Associate. **PADI certified 3x**: Advanced Open Water Diver, Enriched Air Nitrox Diver, Rescue Diver.

## the sea

3rd place, Oceanman 10 km, rio de janeiro, september 2025. 2× the 10 km. strava.